

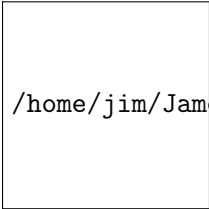
Section 3.2: Basic Terms of Probability

Finite Mathematics MTH 107 Fall 2014

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Mars Hill University

Friday 3rd October, 2014



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Set Theory Notation

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ex. Let $S = \{1, 2, 3, 4, 5, 6, 7\}$ be the universal set of E . Then
 $E' = \{1, 2\}$

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Experiment: a process by which an outcome is obtained.

Sample space: set S of all possible outcomes of an experiment.

Event: any subset $E \subseteq S$.

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Probability of an event, \$\$\$

The *probability* of event E , denoted $p(E)$, is

if the experiment's outcomes are equally likely.

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A die is rolled. Find:

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- e. The probability of rolling below 4

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- f. $o(E_2) = 3 : 3 = 1 : 1$

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Relative Frequency

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Example 3.2.1.

If a student is randomly selected from the class, what is the probability that a female is selected?

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Example 3.2.2.

A baseball player has 2,865 hits in 8,239 times at bat. What are the chances of him getting a hit at his next attempt?

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Law of Large Numbers

If an experiment is repeated a large number of times, the relative frequency of an outcome will become close to the actual probability of that outcome.

Relative Frequency

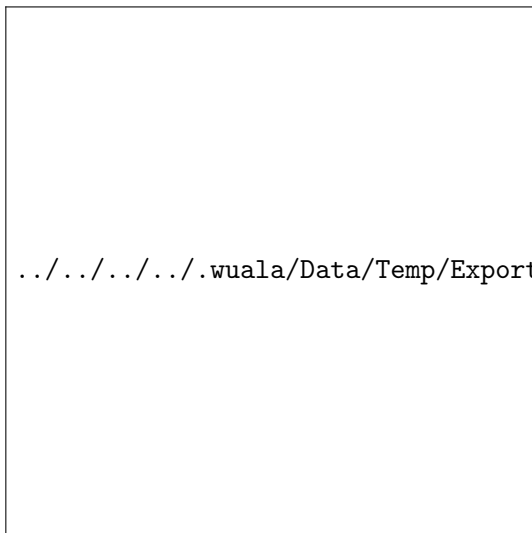


Figure: As the number of rolls increase, the average value approaches 3.5

Problems

Problem 3.2.1. A jar contains 12 black, 8 red, 10 yellow, and 5 green jelly beans. Find the probability of the following:

The probability that you randomly pick a black jelly bean.

Problems

Problem 3.2.2. A jar contains 12 black, 8 red, 10 yellow, and 5 green jelly beans. Find the following:

The odds that you randomly pick a black jelly bean.

Problems

Problem 3.2.3. A jar contains 12 black, 8 red, 10 yellow, and 5 green jelly beans. Find the following:

The probability that you randomly pick a black or red jelly bean.

Problems

Problem 3.2.4. A jar contains 12 black, 8 red, 10 yellow, and 5 green jelly beans. Find the following:

The odds that you randomly pick a black or red jelly bean.

Problem 3.2.5. A jar contains 12 black, 8 red, 10 yellow, and 5 green jelly beans. Find the following:

The probability that you randomly pick a black and red jelly bean.

Problems

Problem 3.2.6. A jar contains 12 black, 8 red, 10 yellow, and 5 green jelly beans. Find the following:

The odds that you randomly pick a jelly bean that is not green.

Problems

Problem 3.2.7. A jar contains 12 black, 8 red, 10 yellow, and 5 green jelly beans. Find the following:

You randomly pick a jelly bean. What is the sample space?

Problem 3.2.8. Find the following:

What is the probability of flipping a coin 3 times and getting all heads?

Problem 3.2.9. Find the following:

What is the probability of flipping a coin 3 times and getting no heads?

Problem 3.2.10. Find the following:

What is the probability of flipping a coin 3 times and getting two heads?

Problem 3.2.11. Find the following:

What is the probability of flipping a coin 3 times and getting at least two heads?

Problem 3.2.12. Find the following:

What is the probability of flipping a coin 3 times and getting one tail?

Problem 3.2.13. Find the following:

What is the probability of flipping a coin 1 time and getting a tail *or* a head?

Problem 3.2.14. Find the following:

What is the probability of flipping a coin 1 time and getting a tail *and* a head?

Problem 3.2.15. Find the following:

A card is drawn from a 52 card deck. What is the probability of getting a queen?

Problem 3.2.16. Find the following:

A card is drawn from a 52 card deck. What is the probability of getting a queen of hearts?

Problem 3.2.17. Find the following:

A card is drawn from a 52 card deck. What is the probability of getting a queen or a heart?